ECN 591 – Market Design

Course Introduction & Syllabus

Tomas Larroucau

Arizona State University

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Overview

A motivating quote

"Only recently have we economists started to understand enough about how markets work so that we can help in that process."

Alvin E. Roth
2012 Nobel Prize laureate (with Lloyd Shapley)
"for the theory of stable allocations and the practice of market design."

Market Design

Market design is the area of economics where economists analyze and improve the way markets work.

Critical ingredients to a successful design:

- Understanding existing institutions
- Good theory
- Good computational modeling (and tools)
- Well-designed experiments

What is a market?

In economics, whenever there is:

- a demand for something
- a supply for that something

There is a market for that something.

The traditional approach consists of finding an equilibrium price, i.e., a price p such that:

$$\#\text{ units buyers are willing to buy at } p \text{ or more } = \#\text{ units sellers are willing to sell for } p \text{ or less.}$$

Questions

- How do we get the equilibrium price?
- Is the "price recipe" the same for all markets?
- What if the price is not the only parameter driving individuals' decisions?
- What if there's no price? (no monetary transactions between sellers and buyers)
- What if the price is "shared"? (e.g., roommates sharing a rent)

The exact details about how to organize a market do matter: they will affect agents' behavior and ultimately affect:

- Who gets what
- At what price (if there's a price)

The easy case

A market for a very common good (e.g., milk):

- Milk is the same everywhere.
- Large number of buyers and sellers.
- Quantities for demand/supply can be adjusted.
- Sellers (i.e., stores) adjust their prices up/down, depending on the sales. Consumers react to prices.
- After some time, prices stabilize and we get an “equilibrium” price.

The usual supply/demand approach works: no need to be specific about the price mechanism.

More challenging

- College admission: large demand (students) and large supply (colleges)
- Price = tuition
- Looks like the market for milk. So, why don't we have a "market for college admission"?

We're obviously not in equilibrium: top schools accept less than 5% of applicants.

For instance, Harvard could raise tuition until

$$\# \text{ applicants} = \# \text{ seats}.$$

When prices are not enough

- Buyers/sellers do not only care about the good they are buying/selling (education), but also with whom they make transaction:
  - Colleges and students have preferences over each other
  - Tuition is not the only decision variable

Conclusion: price may affect people's decisions, but may not be sufficient to determine the "equilibrium" (or the final allocation). Other forces are at work, and they need to be taken into account.

Even worse

What if there's no monetary transaction at all?

- In almost all countries, selling/buying human organs for transplants is illegal
- Yet, there is a market: a demand (patients) and a supply (live/cadaveric donors)
- Is there a way to organize such markets?

Prices may have adverse effects, too:

- It is legal in some countries to compensate blood/sperm/eggs donors
- Such compensation may prevent donation ("I don't do that for money")

What a market needs to work

If a market "works" it means that there are some trades. For this to occur several conditions must or should be met:

- We need "enough" actors from both sides. This is called **market thickness**.
- Sellers need to meet buyers, and buyers need to meet sellers. Roads, internet, or trade treaties bring thickness
- Also, if a seller can face many different buyers she can have a good knowledge of how the demand looks like.

What a market needs to work (cont'd)

- **Avoid congestion:** Too much thickness can create problems, like too much traffic can create traffic jams.
- **Make the market safe:** actors must be able to make the "right" decision
  - They need to understand the rules and how the market works
  - Avoid feeling ripped off (otherwise they don't want to participate)

How does market design work? – Matching markets

Economics studies how goods and services are exchanged or distributed via a market.

Traditionally, markets determine allocation through prices, which are sufficient statistics to determine who gets what.

But in some cases prices may not be enough to characterize allocations (e.g., college admission, labor market, etc.).

It's also about choosing and being chosen.

Let's consider the extreme case where there's no price (i.e., no monetary transaction).

Matching markets: examples

Matching markets are typical examples of markets where there is no monetary transaction (and thus no price):

- School assignment
- Medical match
- Allocation of dorms
- Assignment of cadets to branches
- Organ transplants
- Allocation of subsidized/public housing
- etc.

Syllabus Overview

Syllabus

Course Description

- This course covers the theory and practice of Market Design, focusing on how to fix market failures through the design of rules and algorithms.
- Market design studies how real-world institutions allocate scarce resources when prices alone are insufficient or infeasible, emphasizing incentives, information, and institutional detail.
- We will explore how poorly designed rules can lead to inefficiency, congestion, or strategic behavior and how well-designed mechanisms can improve welfare, with applications to school choice, labor markets, public housing, and organ transplants.
- The course combines theoretical foundations with empirical analysis and practical applications, including a workshop on using AI tools for research.

Course Information

Instructor:	Tomas Larroucau
Email:	`Tomas.Larroucau@asu.edu`
Office:	CRTVC 412
Office Hours:	By appointment
Class Time:	Monday & Wednesday, 1:35 pm – 4:05 pm
Location:	McCord 250
Teaching Assistant:	Martin Carbajal, `macarba3@asu.edu`

Grading

- **Participation (20%):** Attendance is mandatory. Active participation in class is encouraged.
- **Assignments (30%):** Two assignments (15% each) covering empirical aspects of market design. Upload required deliverables to Canvas no later than one week before the next class (dates may be adjusted).
- **Midterm (20%):** Wednesday, 18 February (first half of class; 90 minutes). Covers material up to that point.
- **Final Exam (30%):** Wednesday, 4 March (120 minutes). Comprehensive.

Course Schedule

Note: Schedule is subject to change.

Week	Date	Date	Topics
1	Jan 21	Jan 23	**Course Overview & Tools** — Syllabus, AI Workshop, LaTeX Tools
2	Jan 26	Jan 28	**Foundations** — AI Workshop (Cont.), Intro to Market Design, Mechanisms (DA, IA) *First Assignment*
3	Feb 02	Feb 04	**Theory & Applications I** — Centralized Assignment Mechanisms, School Choice *Second Assignment*
4	Feb 09	Feb 11	**Theory & Applications II** — Revealed Preferences, Information Frictions
5	Feb 16	Feb 18	**Econometrics of Market Design** — Midterm Exam, Econometrics of Treatment Effects in CAS
6	Feb 23	Feb 25	**Entry-Level Markets** — NRMP, Teacher Allocation, Licensing
7	Mar 02	Mar 04	**Other Applications** — Public Housing, Organ Transplants, Final Exam

Course Materials

Textbook

- **Market Design: Auctions and Matching** (MIT Press, 2018)
  Guillaume Haeringer. ISBN: 9780262037549

Handbook Chapters

- **Market Design** — Nikhil Agarwal and Eric Budish
- **Handbook of the Economics of Matching** (Vol. 2, 1st ed.) — Yeon-Koo Che, Pierre A. Chiappori, and Bernard Salanié

Software and Use of AI

Software

- We will use Python for simulations and empirical analysis.
- LaTeX is recommended for submitting problem sets.

Use of Generative AI (Generally Permitted Within Guidelines)

- Use of AI tools is welcome (and sometimes required) with appropriate attribution.
- AI tools may be used to brainstorm, draft, edit, and revise.
- Indeed, some assignments will require the use of AI tools.
- I will teach how to properly use them in the first part of the course.
- If you are unsure what is permitted, ask the instructor before submitting work.